

Exp 7

Aim

To understand the Truffle framework, establish a local Ethereum blockchain using Ganache, and compile, deploy, and interact with a Solidity smart contract on the local blockchain.

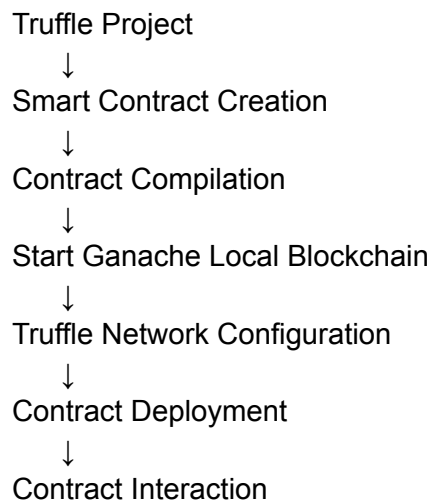
Theory

Truffle is a development framework used for building, compiling, testing, and deploying Ethereum-based smart contracts. It provides a structured project environment containing directories for smart contracts, deployment migrations, tests, and configuration.

Ganache is a local blockchain environment used for Ethereum development and testing. It provides local blockchain accounts with test Ether and allows transactions and smart contracts to be executed without connecting to a public Ethereum network.

In this experiment, Truffle is used to create and configure a blockchain project. A Solidity smart contract named `SimpleStorage` is created to store and retrieve an integer value. The contract is compiled using Truffle and deployed to the local Ganache blockchain through a migration script.

The overall process is:



The smart contract contains two main functions:

- `set(uint256 value)` – stores a value in the blockchain.
- `get()` – retrieves the stored value.

Once deployed, the contract receives a unique contract address on the local blockchain. Transactions generated while deploying and interacting with the contract are recorded as blockchain transactions.

Output / Screenshots

Screenshot 1: Truffle and Ganache Version

```
A ~ >
truffle version
Truffle v5.11.5 (core: 5.11.5)
Ganache v7.9.1
Solidity v0.5.16 (solc-js)
Node v26.8.2
Web3.js v1.10.0

A Desktop/BC_Exp/exp7 >
ganache --version © 12:45
This version of μWS is not compatible with your Node.js build
:

Error: Cannot find module '../binaries/uws_linux_x64_147.node'
'

Require stack:
- /usr/lib/node_modules/ganache/node_modules/@trufflesuite/uw
s-js-unofficial/src/uws.js
- /usr/lib/node_modules/ganache/dist/node/cli.js
Falling back to a NodeJS implementation; performance may be d
egraded.

ganache v7.9.2 (@ganache/cli: 0.10.2, @ganache/core: 0.10.2)
```

Screenshot 2: Truffle Project Initialization

```
truffle init
```

🕒 12:45

```
Starting init...
```

```
=====
```

```
> Copying project files to /home/chirag/Desktop/BC_Exp/exp7
```

```
Init successful, sweet!
```

```
Try our scaffold commands to get started:
```

```
$ truffle create contract YourContractName # scaffold a contract
```

```
$ truffle create test YourTestName          # scaffold a test
```

```
http://trufflesuite.com/docs
```

Screenshot 3: Smart Contract Code

```
GNU nano 9.2 contracts/SimpleStorage.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract SimpleStorage {
    uint256 private storedValue;

    function set(uint256 value) public {
        storedValue = value;
    }

    function get() public view returns (uint256) {
        return storedValue;
    }
}
```

Screenshot 4: Ganache Local Blockchain

```

A ~ >
ganache                                     12:47
This version of μWS is not compatible with your Node.js build:

Error: Cannot find module '../binaries/uws_linux_x64_147.node'
Require stack:
- /usr/lib/node_modules/ganache/node_modules/@trufflesuite/uws-js-unofficial/src/uws.js
- /usr/lib/node_modules/ganache/dist/node/cli.js
Falling back to a NodeJS implementation; performance may be degraded.

ganache v7.9.2 (@ganache/cli: 0.10.2, @ganache/core: 0.10.2)
Starting RPC server

Available Accounts
=====
(0) 0x0c4B1A89B27177b4C657655476566Fb01947C536 (1000 ETH)
(1) 0x9eB26fc2C3100B2fFbDce41D6324CBc3Df193101 (1000 ETH)
(2) 0xCa32938C813a445eA70FceF7Dd17455f2C6aAa06 (1000 ETH)
(3) 0x0CaC1101af17F7CAb51a9A74B46A7a65849619f4 (1000 ETH)
(4) 0xF48765a3e83B950C61f69BFDDed4A10e3e991C87 (1000 ETH)
(5) 0x5f757e4aB79cdC9Ad4555e3580e1F85826B1632c (1000 ETH)
(6) 0x1Ea634ff539B4368343598E221dD8CEa6d5876Ba (1000 ETH)
(7) 0x98c5f0AF685E0dd402554a89F41EFA09011C879e (1000 ETH)
(8) 0x81BBECc5b8127f1bEcA56eC885A589B48ddD2F2d (1000 ETH)
(9) 0x3E309D45989F4e1560DcD02815844be1A2635eBA (1000 ETH)

HD Wallet
=====
Mnemonic:      such enable soon copy genius leader secret already cloth coast scheme main
Base HD Path:  m/44'/60'/0'/0/{account_index}

Default Gas Price
=====
2000000000

BlockGas Limit
=====
30000000

Call Gas Limit
=====
50000000

Chain
=====
Hardfork:  shanghai
Id:        1337

RPC Listening on 127.0.0.1:8545

```

Screenshot 5: Successful Smart Contract Compilation

truffle compile

```
▲ Desktop/BC_Exp/exp7 >
nano truffle-config.js 12:47
▲ Desktop/BC_Exp/exp7 >
nano contracts/SimpleStorage.sol 12:48
▲ Desktop/BC_Exp/exp7 >
nano migrations/2_deploy_contracts.js 12:49
▲ Desktop/BC_Exp/exp7 >
truffle compile 12:49
```

Compiling your contracts...

=====

```
✓ Fetching solc version list from solc-bin. Attempt #1
✓ Downloading compiler. Attempt #1.
> Compiling ./contracts/SimpleStorage.sol
> Artifacts written to /home/chirag/Desktop/BC_Exp/exp7/build
/contracts
> Compiled successfully using:
  - solc: 0.8.21+commit.d9974bed.Emscripten.clang
```

Screenshot 6: Smart Contract Deployment

```
Compiling your contracts...
=====
> Everything is up to date, there is nothing to compile.

Starting migrations...
=====
> Network name:      'development'
> Network id:       1789715829031
> Block gas limit: 30000000 (0x1c9c380)

2_deploy_contracts.js
=====
  Deploying 'SimpleStorage'
  -----
  > transaction hash:      0xa6eaaa8b2bf425cf63d1988407be17bc5
0ee6c5f7878da09670e04fd2dec5243
  > Blocks: 0             Seconds: 0
  > contract address:     0xc7da494f3d3Ee8F82413fdFFe8Ba2D2a
7552F9E
  > block number:        1
  > block timestamp:      1789716046
  > account:             0x0c4B1A89B27177b4C657655476566Fb01
947C536
  > balance:             999.999584885125
  > gas used:             122997 (0x1e075)
  > gas price:            3.375 gwei
  > value sent:           0 ETH
  > total cost:           0.000415114875 ETH

RPC Listening on 127.0.0.1:8545
eth_blockNumber
net_version
eth_accounts
eth_getBlockByNumber
eth_accounts
net_version
eth_getBlockByNumber
eth_getBlockByNumber
net_version
eth_getBlockByNumber
eth_estimateGas
net_version
eth_blockNumber
eth_getBlockByNumber
eth_estimateGas
eth_getBlockByNumber
eth_gasPrice
eth_sendTransaction

Transaction: 0xa6eaaa8b2bf425cf63d1988407be17bc50ee6c5f7878
da09670e04fd2dec5243
Contract created: 0xc7da494f3d3ee8f82413fdfffe8ba2d2a7552f9
e
Gas usage: 122997
Block number: 1
Block time: Fri Sep 18 2026 12:50:46 GMT+0530 (India Stand
ard Time)

eth_getTransactionReceipt
eth_getCode
eth_getTransactionByHash
```

Screenshot 7: Smart Contract Interaction

Screenshot 8: Blockchain Transaction Details

```
Transaction: 0xcb749e180ebb08ac2c020457196278db7cdf96a050bbcb234171d4e3f0d1615e4
Gas usage: 43696
Block number: 2
Block time: Fri Sep 18 2026 12:52:17 GMT+0530 (India Standard Time)
```

Thus, the Truffle framework was successfully used to create and configure a blockchain project. A local Ethereum blockchain was established using Ganache, and the `SimpleStorage` smart contract was successfully compiled, deployed, and tested. The stored value was successfully retrieved from the deployed contract, demonstrating the basic process of smart contract development and deployment on a local blockchain.